



COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

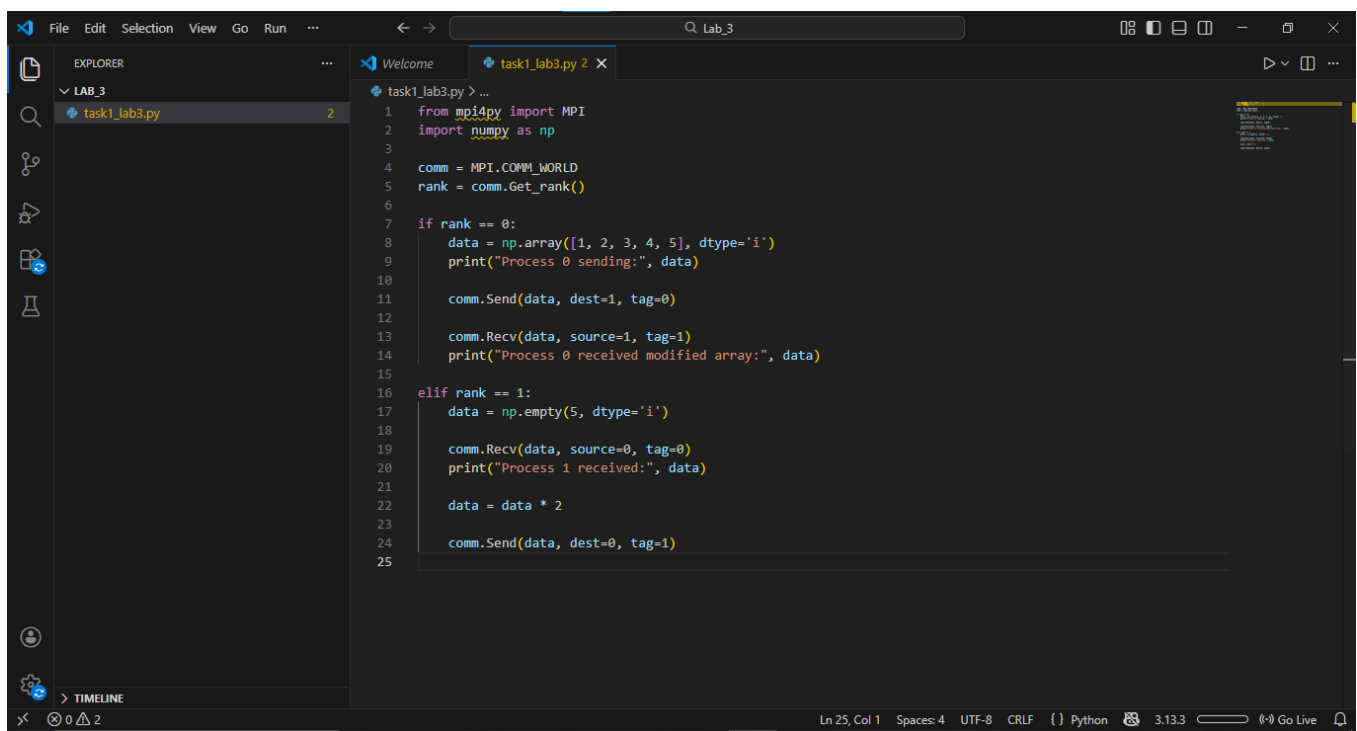
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LAB : 3

Task-1: Write an MPI program where **Process 0** sends an integer array of size 5 to **Process 1**, which then modifies the array (e.g., doubles each value) and sends it back to **Process 0**. Process 0 should then print the modified array.

Code In VS Code



```
1 from mpi4py import MPI
2 import numpy as np
3
4 comm = MPI.COMM_WORLD
5 rank = comm.Get_rank()
6
7 if rank == 0:
8     data = np.array([1, 2, 3, 4, 5], dtype='i')
9     print("Process 0 sending:", data)
10
11     comm.Send(data, dest=1, tag=0)
12
13     comm.Recv(data, source=1, tag=1)
14     print("Process 0 received modified array:", data)
15
16 elif rank == 1:
17     data = np.empty(5, dtype='i')
18
19     comm.Recv(data, source=0, tag=0)
20     print("Process 1 received:", data)
21
22     data = data * 2
23
24     comm.Send(data, dest=0, tag=1)
25
```

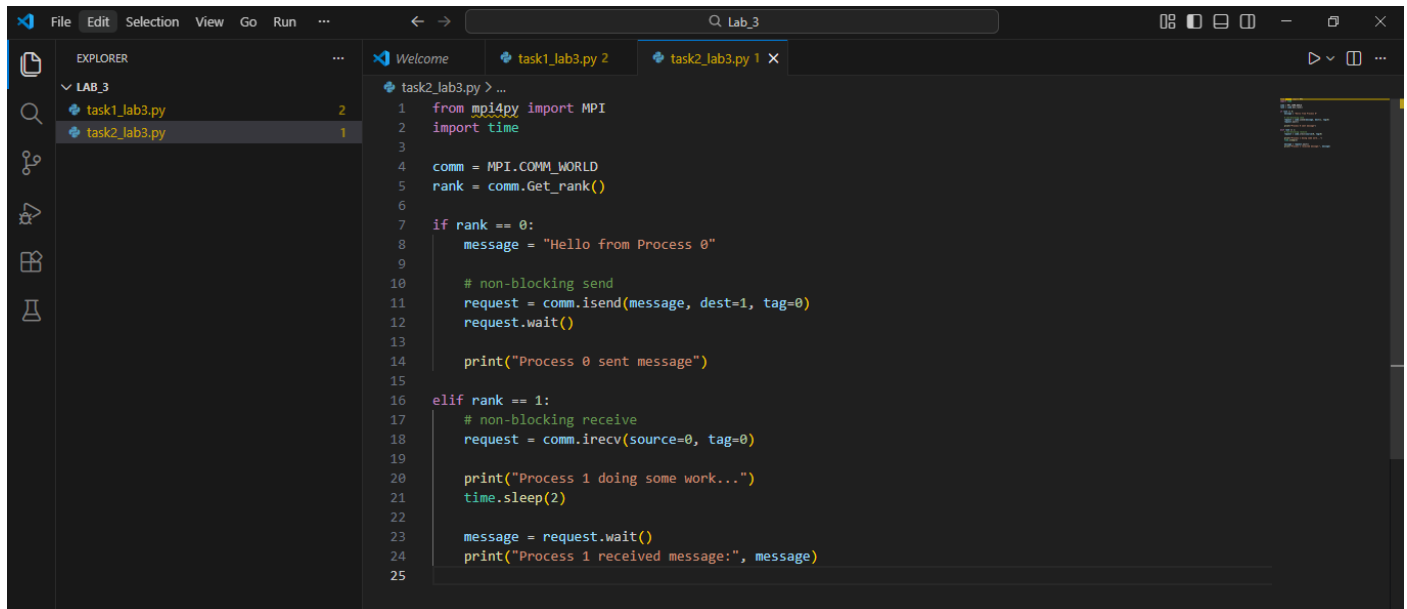
Output (In Anaconda Prompt)

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_3>mpiexec -n 2 python task1_lab3.py
Process 1 received: [1 2 3 4 5]
Process 0 sending: [1 2 3 4 5]
Process 0 received modified array: [ 2  4  6  8 10]

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_3>
```

Task-2: Write an MPI program where **Process 0** sends a string message to **Process 1** using non-blocking communication (isend and irecv). Process 1 should receive the message and print it while performing another computation in parallel.

Code In VS Code



```
1 from mpi4py import MPI
2 import time
3
4 comm = MPI.COMM_WORLD
5 rank = comm.Get_rank()
6
7 if rank == 0:
8     message = "Hello from Process 0"
9
10    # non-blocking send
11    request = comm.isend(message, dest=1, tag=0)
12    request.wait()
13
14    print("Process 0 sent message")
15
16 elif rank == 1:
17    # non-blocking receive
18    request = comm.irecv(source=0, tag=0)
19
20    print("Process 1 doing some work...")
21    time.sleep(2)
22
23    message = request.wait()
24    print("Process 1 received message:", message)
25
```

Output (In Anaconda Prompt)

```
(mpi_env) C:\Users\Rifaa Ajmal\Desktop\Lab_3>mpiexec -n 2 python task2_lab3.py
Process 0 sent message

Process 1 doing some work...
Process 1 received message: Hello from Process 0

(mpi_env) C:\Users\Rifaa Ajmal\Desktop\Lab_3>
```